

ICMP Protocol Architecture and Transfer Design

Recommended Architecture

Function	Receiver Recommendation
Port knock	Scapy AsyncSniffer
Connect/disconnect command	Scapy AsyncSniffer
Transfer-file request	Scapy AsyncSniffer
Run-program request	Scapy AsyncSniffer
ACK_READY packet	Raw socket
File chunk packets	Raw socket
Chunk ACK packets	Raw socket
Program command stream	Raw socket
Program output stream	Raw socket
Program output ACKs	Raw socket

Control Plane vs Data Plane

Keep Scapy AsyncSniffer for low-volume control packets such as port knocks and requests. Use raw sockets for high-volume file chunks, command streams, output streams and ACK packets.

Option A: 2-byte sequence + 2-byte data

- 1 Read file into bytes.
- 2 Split into chunks of 1024 packets (2048 bytes per chunk).
- 3 Send START packet with filename, file size and total chunks.
- 4 Send packets with sequence 0-1023.
- 5 Receiver stores packets by sequence.
- 6 Receiver sends ACK only after the whole chunk is complete.
- 7 Sender waits for ACK before sending next chunk.
- 8 Resend same chunk on timeout.
- 9 Send END packet after the final chunk.

Option B: 1-byte sequence + 3-byte data

- 1 Read file into bytes.
- 2 Split into chunks of 256 packets (768 bytes per chunk).
- 3 Send START packet.
- 4 Send packets with sequence 0-255.
- 5 Receiver stores packets by sequence.
- 6 Receiver ACKs only when the full chunk is present.

- 7 Sender waits for ACK before sending next chunk.
- 8 Resend chunk on timeout.
- 9 Send END packet.